==

INTRUSION DETECTION SYSTEM

Network Malicious Detection (Benign vs Malicious)

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All IDSAI Dataset shape: (1000000, 26)

Dataset after drop duplicates: (693116, 26)

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Label (Benign and Malicious) counts:

label

1   471256

0   221860

Name: count, dtype: int64

Features after cleaning: 21

Data split: Train=485181 (70%), Test=207935 (30%)

Training model...

Training time: 132.98 seconds

Cross-Validation (on training set)...

10-Fold CV Accuracy on Train: 0.9994 ± 0.0001

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PERFORMANCE SUMMARY

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Accuracy | Train: 99.95% | Test: 99.95%

Balanced Acc | Train: 99.94% | Test: 99.94%

Precision | Train: 99.95% | Test: 99.95%

Recall | Train: 99.95% | Test: 99.95%

F1 | Train: 99.95% | Test: 99.95%

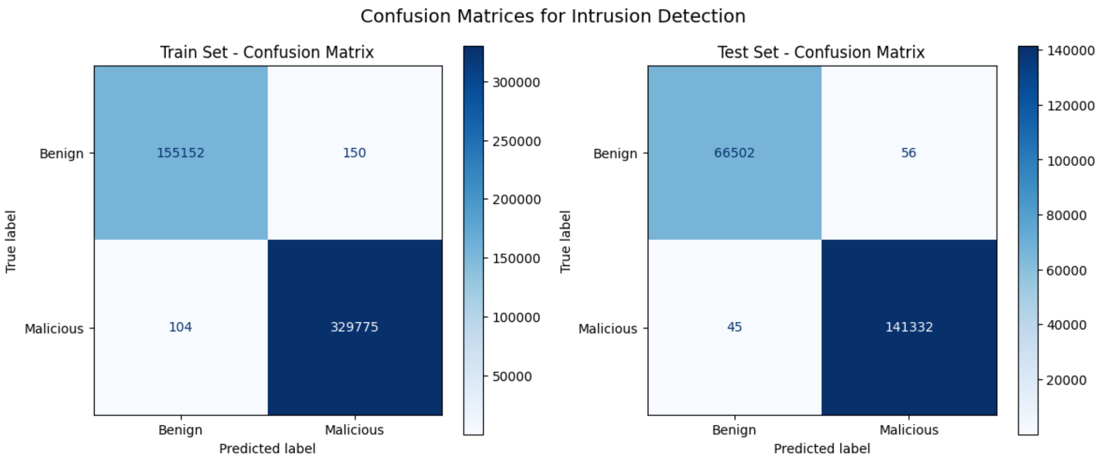
MCC | Train: 99.88% | Test: 99.89%

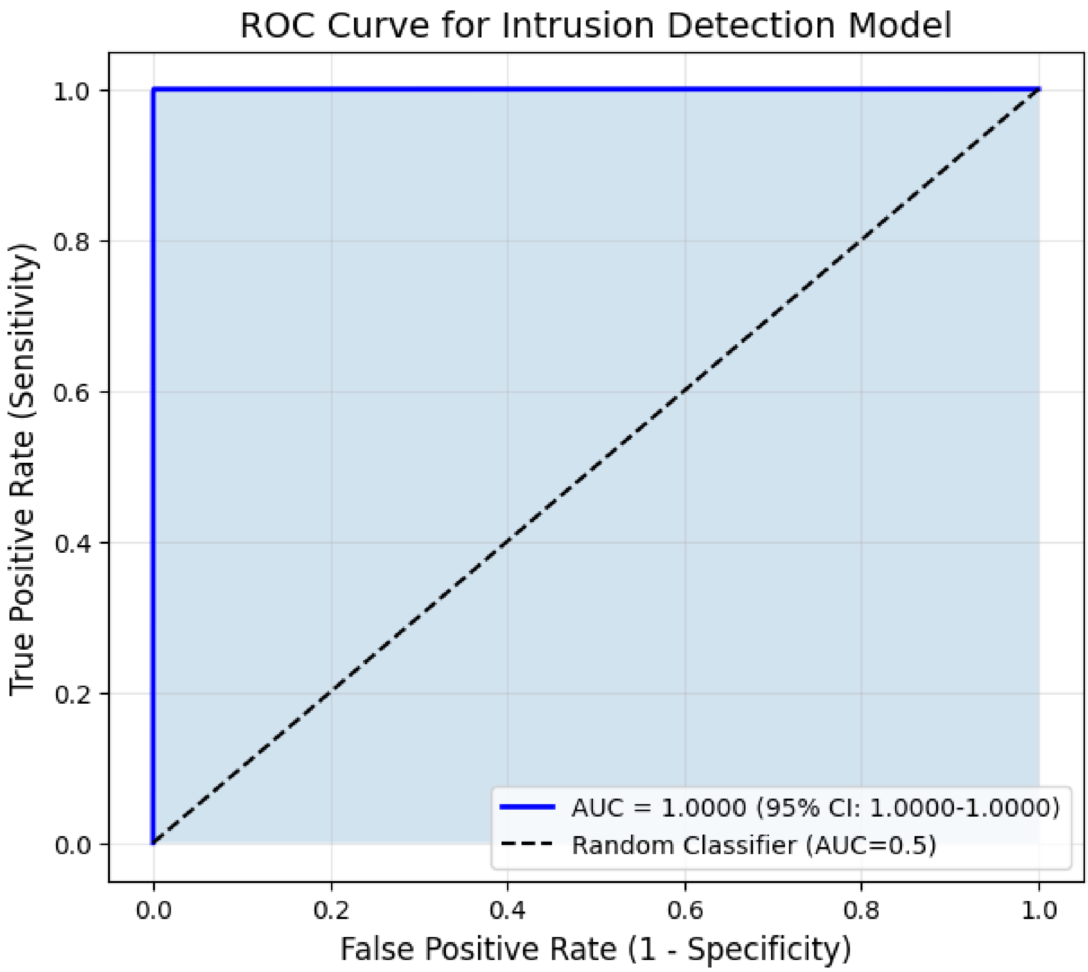
Kappa | Train: 99.88% | Test: 99.89%

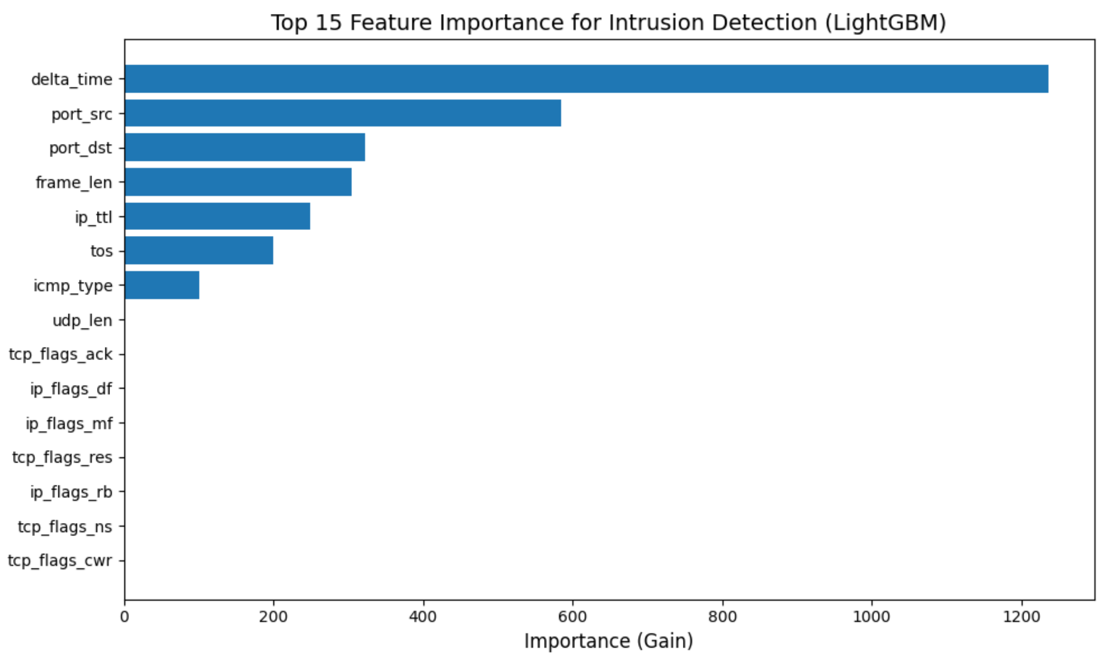
Specificity | Train: 99.90% | Test: 99.92%

Sensitivity | Train: 99.97% | Test: 99.97%

AUC | Train: 0.00% | Test: 100.00%







Top 10 features:

Feature Importance

0	delta_time	1236
1	port_src	585
2	port_dst	322
3	frame_len	304
5	ip_ttl	249
7	tos	200
6	icmp_type	100
4	udp_len	2
16	tcp_flags_ack	2
9	ip_flags_df	0

SHAP analysis (sampling 100 test samples)...

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FEATURE IMPORTANCE BASED ON SHAP

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Feature	Strength	Impact %
frame_len	3.2581	47.41
port_dst	1.3421	19.53
tos	0.9327	13.57
port_src	0.7628	11.10
ip_ttl	0.3880	5.65
icmp_type	0.1389	2.02
delta_time	0.0479	0.70
tcp_flags_ack	0.0019	0.03
udp_len	0.0000	0.00

ip_flags_df	0.0000	0.00
ip_flags_mf	0.0000	0.00
tcp_flags_res	0.0000	0.00
ip_flags_rb	0.0000	0.00
tcp_flags_ns	0.0000	0.00
tcp_flags_cwr	0.0000	0.00

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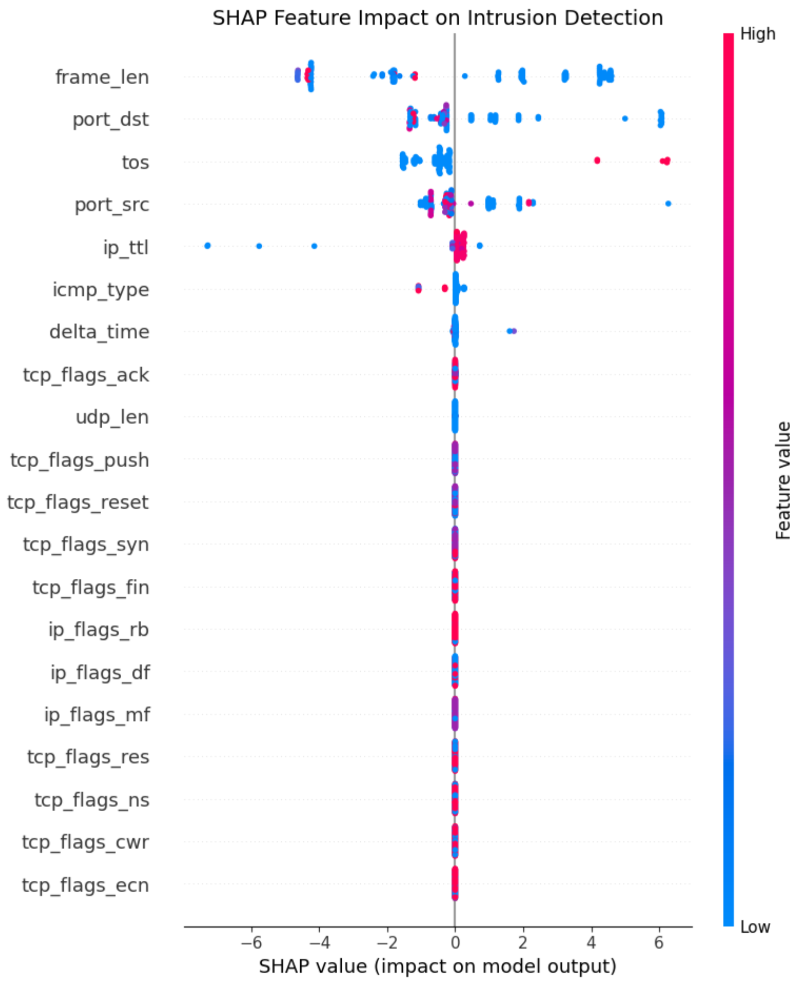
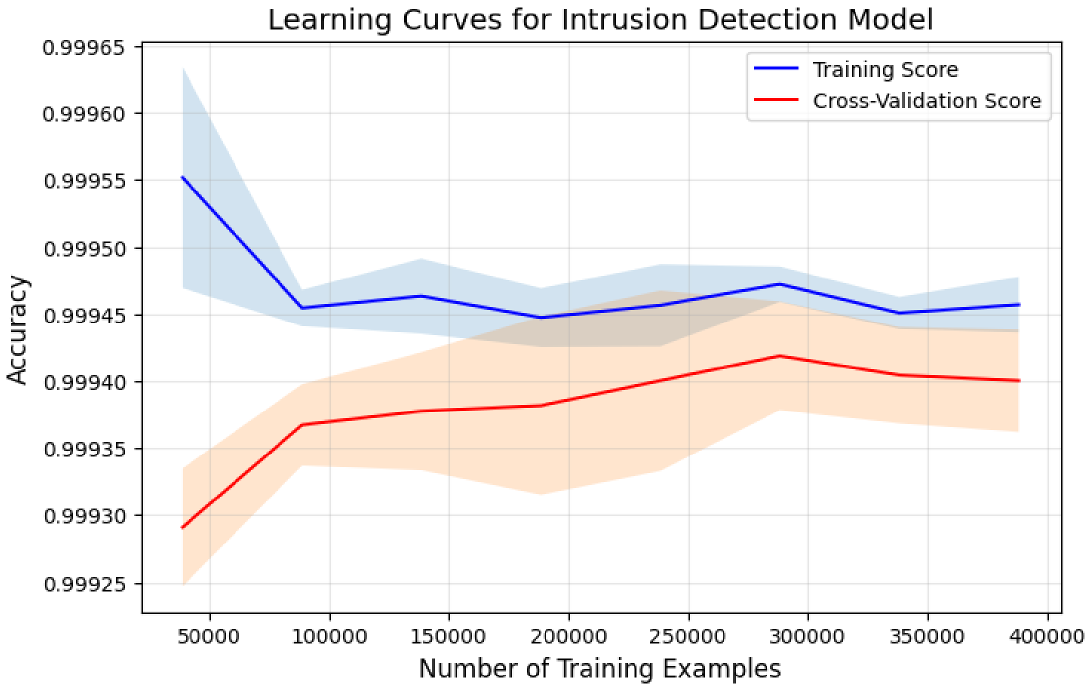
Top feature: frame_len (Impact: 47.41%)

Second feature: port_dst (Impact: 19.53%)

Third feature: tos (Impact: 13.57%)





Comparing with individual models on test set...

	Accuracy	Balanced Acc	Precision	Recall	F1	MCC	Kappa \
xgb	0.9995	0.9994	0.9995	0.9995	0.9995	0.9989	0.9989
Ensemble	0.9995	0.9994	0.9995	0.9995	0.9995	0.9989	0.9989
lgb	0.9995	0.9994	0.9995	0.9995	0.9995	0.9988	0.9988

	Specificity	Sensitivity	AUC
xgb	0.9991	0.9997	1.0
Ensemble	0.9992	0.9997	1.0
lgb	0.9993	0.9996	1.0

McNemar's test (Ensemble vs Logistic Regression): p-value = 0.000000

Result: Statistically significant difference (p < 0.05)

Model saved to fast_ids_model.pkl

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FINAL CLASSIFICATION REPORT (Test Set)

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	precision	recall	f1-score	support
Benign	1.00	1.00	1.00	66558
Malicious	1.00	1.00	1.00	141377
accuracy			1.00	207935
macro avg	1.00	1.00	1.00	207935
weighted avg	1.00	1.00	1.00	207935

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INTRUSION DETECTION SYSTEM - COMPLETED SUCCESSFULLY
Results saved in: ids_results/
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Sample predictions on test instances:
  Correct | True: Benign    | Predicted: Benign
  Correct | True: Malicious | Predicted: Malicious
  Correct | True: Malicious | Predicted: Malicious
  Correct | True: Benign    | Predicted: Benign
  Correct | True: Benign    | Predicted: Benign

Total execution time: 14.17 minutes
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